

vibechecker — Project Progress

Maps client meetings, stated requirements, and development phases from `VibeGui Project.md` to git commits. Useful for tracking what has been delivered and what remains.

Project Background

Originated 2025-01-23 when a Digiducer-based data collection script was demonstrated to Carmen at a client site. The goal: replace expensive commercial vibration analysis tools (e.g. VibeAnalyze, \$999) with a purpose-built Python desktop app targeting industrial predictive maintenance workflows.

Development Phases

Proto — Initial prototype (Feb – Aug 2025)

Bare-bones data capture and display using the Digiducer USB audio interface. No HDF5, no velocity spectrum, minimal GUI.

Date	Commit	Description
2025-02-15	<code>ae1736e</code>	Project created
2025-03-05	<code>5b017d1</code>	Project management notes
2025-03-17	<code>891d26f</code>	Backup snapshot
2025-05-20	<code>cca1d4e</code>	Working demo (Digiducer stream + basic plot)
2025-05-20	<code>e247dba</code>	Module init
2025-05-21	<code>856f855</code>	Digiducer abstracted into sensor and logger types
2025-08-12	<code>1257994</code>	First working dearpygui GUI with live stream
2025-08-12	<code>a3c2fd1</code>	Project management
2025-08-12	<code>be9e91c</code>	Bug fix: array dimensions

Meeting — 2025-01-23: Carmen, first demo

Requirements captured:

#	Requirement	Status
R1	Frequency range and bin size selector	✓ Done — Dec 2025
R2	Normalize FFT to IPS or g (velocity preferred over acceleration)	✓ Done — Dec 2025
R3	Display top peaks	✓ Done — Dec 2025
R4	Smoothing	✗ Not implemented
R5	Interactive peak inspection	✗ Not implemented
R6	Running speed harmonics markers	✗ Not implemented

#	Requirement	Status
R7	Bearing fault frequency markers	✗ Not implemented
R8	Metadata: date, company, machine info, bearing types, running rate	✗ Not implemented
R9	Trend: overall vibration over time	✓ Done — Mar 2026
R10	Luxury: startup and coastdown trend	✗ Future

Phase 0 — Digiducer acquisition, GUI polish (Nov – Dec 2025)

Core pipeline stabilised; units, HDF5 persistence, and modular architecture added.

Date	Commit	Description
2025-11-17	587e393	Save button added
2025-11-20	e9120d2	Logging module with YAML config; print statements removed
2025-11-25	c8bb0ae	VibeSample refactored
2025-11-26	6049f7f	Carmen meeting notes — deployment roadmap
2025-12-11	22014af	Units switching; save/load via pickle
2025-12-16	10e48a0	scipy FFT tools; GUI unit and integration controls
2025-12-17	580d254	Logger made universal
2025-12-17	383f28e	File browser fix
2025-12-17	14dd5ca	Unit conversion consistency
2025-12-17–18	772214e – 4424d3d	AcquisitionSettings upgrade (multi-step)
2025-12-29	38f22d3	Plot units upgrade — units switching throughout GUI
2025-12-29	8fb488c	refactor_util — split into sample, sensor, simulation, digiducer modules

Meeting — 2025-11-25: Carmen, deployment roadmap

Phase definitions agreed:

Phase	Scope	Budget
Phase 1	Development to date (Digiducer era)	\$2k
Phase 2	Polish and deploy to Windows	\$1k
Phase 2b	Integrate BNC DAQ for proximity probes	\$1k
Phase 3	Sales	—

Requirements from this meeting:

#	Requirement	Status
R11	Switch between velocity and acceleration in frequency domain	✓ Done — Dec 2025 (10e48a0 , 38f22d3)

#	Requirement	Status
R12	IPS zero-to-peak trend analysis	✓ Done — Jan 2026, extended Mar 2026
R13	RMS amplitude option	✓ Done — Mar 2026 (5872cab , 1e040d3)
R14	Two simultaneous sensors (phasing)	✓ Done — Mar 2026 (multi-channel pipeline)
R15	Proximity probe / DAQ support	↩ Partially done — PicoScope replaces BNC DAQ

Meeting — 2025-12-30: Carmen

Requirements from this meeting:

#	Requirement	Status
R16	Overall vibration energy (IPS 0-P)	✓ Done — Jan 2026 (ff4b31d)
R17	Trend plot of overall vibration over time	✓ Done — Mar 2026 (cff702e)
R18	Fix IPS unit scaling (was ~200× wrong)	✓ Done — Dec 2025 / confirmed Jan 2026 (e7eddde)

Phase 1b — HDF5, FFT validation, overall vibration (Jan 2026)

Date	Commit	Description
2026-01-05	dc64e37	UI_Elements moved to util for global access
2026-01-05	e3a988d	AcquisitionSettings hot-swap while streaming
2026-01-05	cde0834	Reorg and cleanup
2026-01-06	32ab828	HDF5 save/load (replaces pickle)
2026-01-09	e76fb4f	DAQ hardware evaluation notes committed
2026-01-10	f0c549e	ISO timestamp; monotonic rel_time per block
2026-01-11	e7eddde	FFT amplitude scaling validated with single-tone tests
2026-01-12	ff4b31d	Overall 0-P vibration display (acceleration and velocity)
2026-01-12	fcf9e7e	Oversample stream for better Welch FFT resolution
2026-01-16	70c51db	uldaq dependency added (MCC USB-1608FS-Plus evaluation)
2026-01-20	2f95891	MCC DAQ code examples
2026-03-09	2508756	MCC DAQ integration work

Meeting — 2026-01-11

Notes: - Overall O-P vibration display added and demonstrated - Bin selector behaves poorly — flagged - Timestamp now FAT32-compatible (colons replaced by hyphens) - UI/UX language and jargon should match Alta's tooling - Demo screenshots needed for website

#	Requirement	Status
R19	Confirm overall vibration values are correct (vs. individual peaks)	✅ Done — validated <code>e7eddde</code>
R20	Fix bin selector behaviour	✅ Done — Mar 2026 (<code>a69f14f</code> , <code>3fde4bc</code>)
R21	UI/UX language aligned with Alta	❌ Ongoing
R22	Demo screenshots for website	❌ Not committed

Phase 2 — PicoScope 4000A acquisition backend (Mar 2026)

Hardware pivot: replace Digiducer/sounddevice with PicoScope 4000A. Enables any IEPE sensor (accelerometers, proximity probes, microphones) via standard BNC input — removes Digiducer hardware dependency.

Date	Commit	Description
2026-03-13	<code>cf89848</code>	PicoScope 4000A acquisition backend — <code>FindPicoScope()</code> , <code>PicoScopeStream</code> , ADC → mV, accumulator, USB power fallback
2026-03-13	<code>1b711c9</code>	Merge feature/picoscope into main
2026-03-13	<code>2b9aa38</code> <code>d9add87</code>	Comprehensive README added
2026-03-13	<code>91b06ce</code>	CLAUDE.md architecture guide added
2026-03-16	<code>8abb72d</code>	IEPE sensor registry, triax support, PicoScope GUI integration — <code>ScopeSensor</code> dataclass, <code>ScopeSensorRegistry</code> YAML persistence, per-channel sensitivity (mV/EU), GUI sensor library panel
2026-03-16	<code>cd40f63</code>	Harden PicoScope detection: overvoltage recovery, watchdog (5 s timeout, 3 reconnect attempts)
2026-03-16	<code>f03255d</code>	Fix handle-close between retries; extended delay after <code>PICO_NOT_RESPONDING</code>
2026-03-16	<code>67ebfd6</code>	Test: update for new default voltage range

Phase 2a — Multi-channel EU pipeline and GUI overhaul (Mar 2026)

Full multi-channel data path from hardware to display, with per-channel sensor assignment, modality-aware integration, and three-column GUI layout.

Date	Commit	Description
2026-03-18	<code>0d8911c</code>	Three-column GUI layout with primary window resizing
2026-03-18	<code>0147231</code>	Modality-aware integration (acc ↔ vel ↔ disp via FFT); mV/ EU sensitivity pipeline; unit conversion tests

Date	Commit	Description
2026-03-19	13b1095	Add mil (1/1000 in) as displacement unit
2026-03-20	afa5ce0	Fix frequency-domain integration using signed exponent
2026-03-24	09cea74	Multi-channel pipeline, scope sensor registry, EU unit refactor — PicoScopeStream multi-channel buffers; per-channel mV → EU via ScopeSensor; DataCollector fan-out
2026-03-24	4f8897c	GUI overhaul — channel config panel; per-channel status lights; dual-axis plots; scope sensor registry UI
2026-03-24	8280bd5	Test restructure: multi-channel, hardware isolation, new API coverage
2026-03-24	5d57ab4	Fix PicoScope circular buffer wraparound in streaming callback
2026-03-24	0ca1251	Replace Acquire/Configure tabs with streamlined left-panel layout
2026-03-24	c70b8f3	Visual section containers (borders, themed backgrounds) for left panel
2026-03-25	cff702e	Multi-channel data pipeline — 32-frame ring cache; ChannelResult frozen dataclass; trend store; multi-channel HDF5 layout; browse ← → buttons
2026-03-25	142dc30	Expanded multi-channel test coverage; TestFrameCache, TestTrend, TestChannelResult
2026-03-25	32410b6	Fix: overflow bitmask; default voltage range 10 → 7 (10× finer ADC resolution)
2026-03-25	23d0902	Test: overflow key handling, per-channel helpers
2026-03-25	e365099	Per-channel AC/DC coupling; results card sizing; 3 default peaks
2026-03-25	5ec3e45	Fix: reset channel config and plot series on device switch

Phase 2b — Signal generator, spectrum controls, GUI polish (Mar 2026)

AWG output for excitation testing; FFT window selection; manual axis control; left-panel redesign; siggen config persistence.

Date	Commit	Description
2026-03-25	f74dd41	Signal generator UI — Generate config tab (enable, waveform, freq, amplitude, offset); SimulatedSensor buried (CI/test only via VibeSensor.simulated())
2026-03-25	2a05ff9	FFT window type selector (Hann/Blackman-Harris/Flattop/Hamming/Boxcar/Bartlett)
2026-03-25	7e6b89f	Manual axis control — remove auto-fit on redraw; Autoscale button; Clear Cache button; full Welch frequency range (remove maxfreq display crop)
2026-03-25	ffb4121	Left panel rework — Device card, Channels card, button visibility fix (width=-1 bug), Load button restored

Date	Commit	Description
2026-03-25	83a980d	Card heights (DPG autosize_y fix); toggle button colour theming (idle/waiting/active); autoscale on first acquisition
2026-03-25	ef0b560	Channel Setup and Signal Generator buttons in Channels card
2026-03-25	d0c1068	Device card expanded with model, S/N, ID, channel count
2026-03-25	7bc729d	CH_WARNINGS_SECTION height scales to overflow channel count
2026-03-25	eb811b6	Siggen config persisted to <code>channel_assignments.yaml</code> ; restored on device connect
2026-03-25	7aa69b3	Unified VibeSample pipeline — <code>process()</code> replaces <code>get_accel()</code> + <code>fft()</code> ; returns <code>ChannelResult</code> ; cross-modality FFT integration for physically correct time-domain signal
2026-03-25	5872cab	Amplitude mode selector (RMS / 0-P / P-P) in GUI; Y-axis labels per mode
2026-03-25	1e040d3	Amplitude mode moved to <code>ScopeSensor</code> (per-sensor, not global spectrum setting)
2026-03-26	a69f14f	Modernised AcquisitionSettings — derived properties (samplerate, blocksize from maxfreq/binsize); explicit highpass/lowpass filter fields; Welch overlap; <code>n_fft_bins</code> ; <code>memory_bytes</code>
2026-03-26	ad98816	Lowpass filter added to collector pipeline
2026-03-26	3fde4bc	Spectrum tab redesign — combo presets; live-updating derived fields; filter controls
2026-03-26	36db391	Tests updated for new AcquisitionSettings API

Meeting — 2026-03-18

#	Requirement	Status
R23	Multi-channel: toggle channels on/off	✅ Done (09cea74 , 4f8897c)
R24	Two-channel phase relationship display	❌ Not implemented
R25	Time-domain zoom window (e.g. 5 ms)	❌ Not implemented
R26	Track down velocity integration low-frequency noise	🔄 Partially addressed (afa5ce0); warrants further investigation
R27	Windows packaging (exe)	❌ Not implemented

Meeting — 2026-03-25

#	Requirement	Status
R28	0-P, P-P, RMS selectable per sensor	✅ Done (5872cab , 1e040d3)

#	Requirement	Status
R29	Acquisition settings causing long capture times	✓ Done — AcquisitionSettings now derives samplerate / blocksize from maxfreq / binsize with correct power-of-2 arithmetic (a69f14f)
R30	Displaying wrong bins in spectrum setup	✓ Done — Spectrum tab redesigned with live derived fields (3fde4bc)

Open Requirements / Backlog

Items captured from meetings or todo lists that have not yet been implemented:

#	Source	Requirement
R4	Jan 2025	Spectral smoothing
R5	Jan 2025	Interactive peak inspection (click peak → identify frequency)
R6	Jan 2025	Running speed harmonic markers on spectrum
R7	Jan 2025	Bearing fault frequency markers on spectrum
R8	Jan 2025	Measurement metadata (company, machine, bearing types, running rate)
R10	Jan 2025	Startup / coastdown trend (luxury)
R21	Jan 2026	UI/UX language aligned with Alta tooling
R22	Jan 2026	Demo screenshots for website
R24	Mar 2026	Two-channel phase relationship display
R25	Mar 2026	Time-domain zoom window
R26	Mar 2026	Velocity integration low-frequency noise investigation
R27	Mar 2026	Windows packaging (exe)
—	Future	Sexy web portal for data sharing
—	Future	Proximity probe / DAQ integration (prox probes need only one integration step)

Architecture Evolution Summary

Era	Hardware	Key modules	Format
Proto (Feb–Aug 2025)	Digiducer USB audio	monolithic	—
Phase 0 (Nov–Dec 2025)	Digiducer	sensor, sample, simulation, digiducer	pickle

Era	Hardware	Key modules	Format
Phase 1b (Jan 2026)	Digiducer	+ HDF5, refactored AcquisitionSettings	HDF5
Phase 2 (Mar 2026)	PicoScope 4000A	+ picoscope, scope_sensor, scope_sensor_registry	HDF5 multi-channel
Phase 2a (Mar 2026)	PicoScope	+ ChannelResult, multi-channel pipeline, trend	HDF5 multi-channel
Phase 2b (Mar 2026)	PicoScope + AWG	+ signal generator, Welch controls, derived AcquisitionSettings	HDF5 + YAML config